

STUDENT BRIEFING

Autonomous Robotic Car Challenge

Five specialised subsystems. One integrated mission.

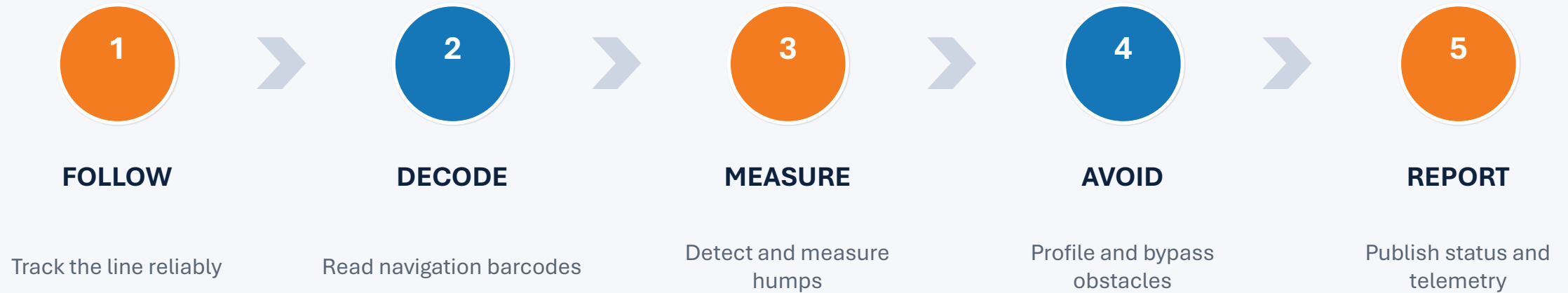
Embedded Systems Project

Design • Test • Integrate • Demonstrate



The Challenge

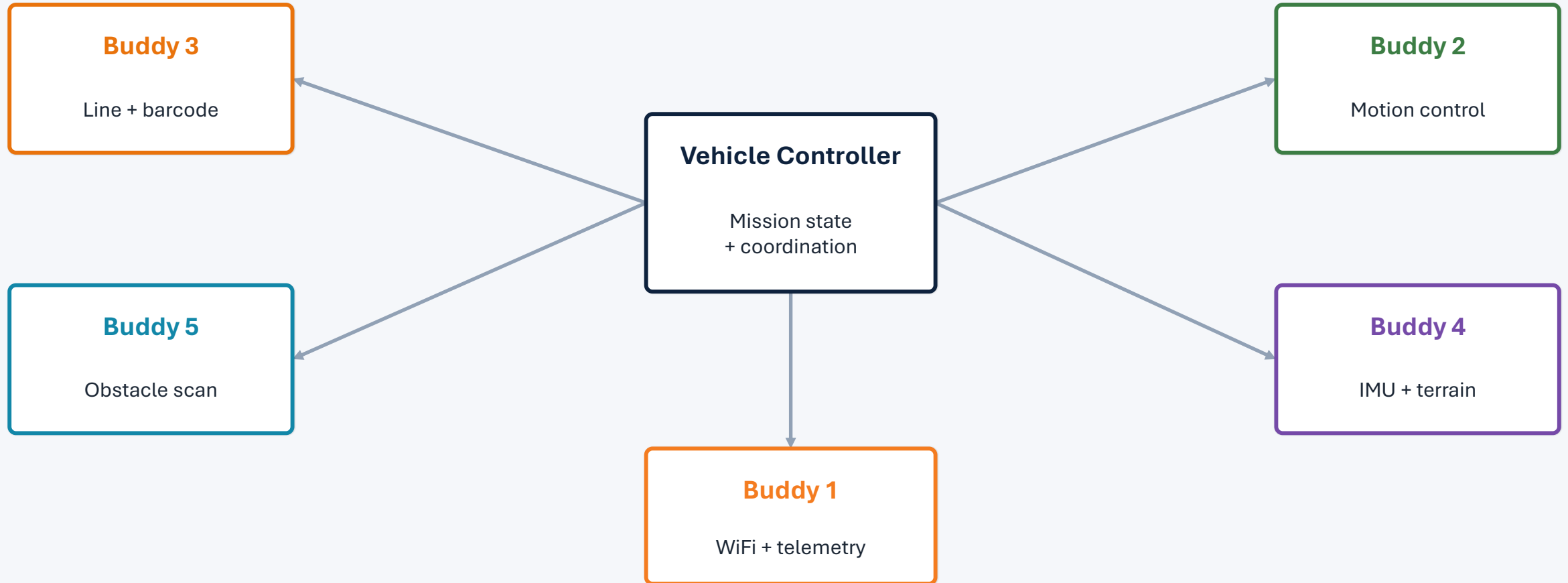
Build an autonomous robotic car that senses, decides, moves and reports as one dependable system.



Success depends on clean interfaces, repeatable subsystem tests and disciplined integration.

How the Five Buddies Fit Together

Each buddy owns a subsystem, but the robot succeeds only when data and commands flow cleanly.



Buddy 1: WiFi Communication, Command and Telemetry

Primary hardware: WiFi



Core work

- Establish WiFi and MQTT
- Design topics and message structures
- Publish telemetry and subscribe to commands
- Recover connections and report heartbeat

Expected deliverables

- Communication API
- Telemetry framework
- MQTT documentation
- Telemetry demonstration

Buddy 2: Motion Control System

Primary hardware: Motors + wheel encoders



Core work

- Integrate motor drivers and encoders
- Estimate speed and distance
- Tune PID speed control
- Correct straight motion and execute turns

Expected deliverables

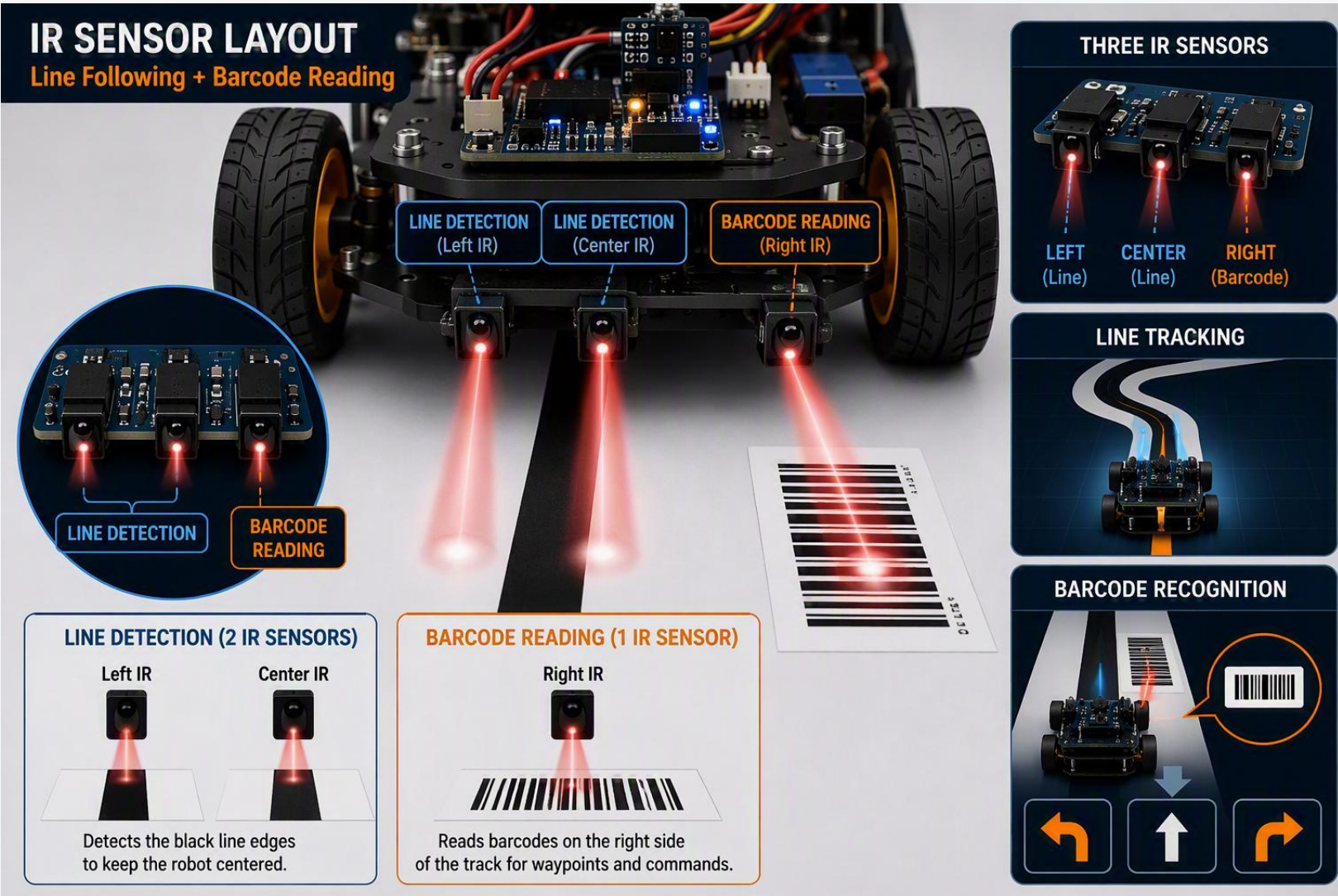
- Motion-control library
- PID tuning report
- Motion accuracy evaluation

Buddy 3: Barcode Decoding and IR Line Following

Primary hardware: 3 × IR sensors

IR SENSOR LAYOUT

Line Following + Barcode Reading



Core work

- Calibrate sensors and estimate line position
- Follow lines and detect junctions
- Detect and decode barcodes
- Generate navigation commands

Expected deliverables

- Line-following module
- Barcode decoder
- Navigation command interface

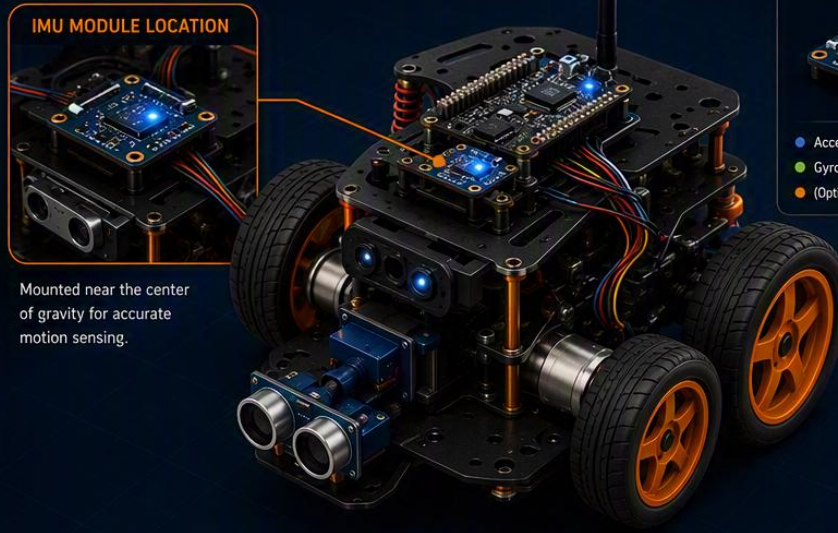
Buddy 4: IMU-Based Motion and Terrain Monitoring

Primary hardware: IMU

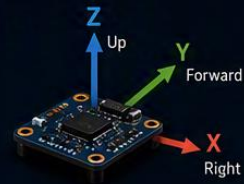
IMU – MOTION & TERRAIN MONITORING

Detect shock, bumps (and measure the height) and direction.

IMU MODULE LOCATION



IMU AXES (RIGHT-HANDED FRAME)



- Accelerometer : Measures linear motion (g)
- Gyroscope : Measures rotation (°/s)
- (Option) Magnetometer : Measures heading

WHAT THE IMU DETECTS



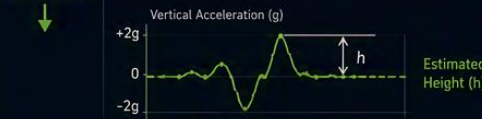
1. SHOCK DETECTION

Detects sudden, high-magnitude impacts.



2. BUMP DETECTION & HEIGHT MEASUREMENT

Detects bumps and estimates their height from vertical acceleration.



Approximate height: $h \approx \frac{v^2}{2g}$ (from vertical acceleration impulse)
 v = vertical velocity change, $g = 9.81 \text{ m/s}^2$



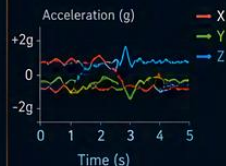
3. DIRECTION & ORIENTATION

Measures vehicle orientation and changes in direction.



REAL-TIME MONITORING

Live data stream to onboard controller.



TERRAIN PROFILING

Build a profile of the track for smarter navigation.



EVENT REPORTING

Log and report motion events.

- ⚠ Shock detected
- ↑ Bump: 3.2 cm
- ➡ Turn right
- ↩ Turn left

INTEGRATION

Use data to adapt speed, reroute, or alert.

- ⚙ Slow down on rough terrain
- ➡ Reroute around hazards
- ⚠ Alert via telemetry

OUTPUTS FOR THE SYSTEM



Accurate IMU placement + calibration = reliable detection of shocks, bumps (and their height) and direction.

Core work

Calibrate the IMU

Detect tilt, humps and peak height

Classify motion and detect collision

Report turn rate and motion events

Expected deliverables

IMU processing module

Hump-detection algorithm

Terrain analysis report

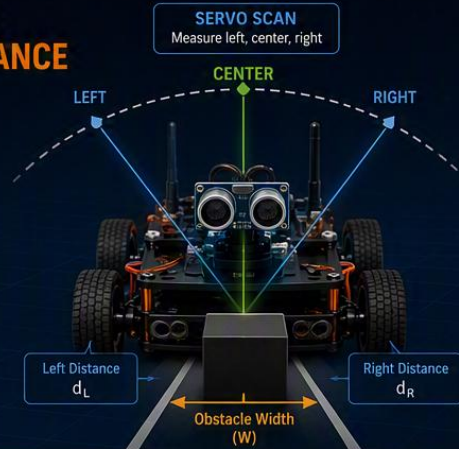
Buddy 5: Adaptive Ultrasonic Scanning and Obstacle Profiling

Primary hardware: Ultrasonic sensor + servo

ADAPTIVE OBSTACLE DETECTION & AVOIDANCE

Detect an obstacle on its path, measure its width using ultrasonic + servo, choose left/right to go (off line) and collect back to the line behind the obstacle.

HARDWARE



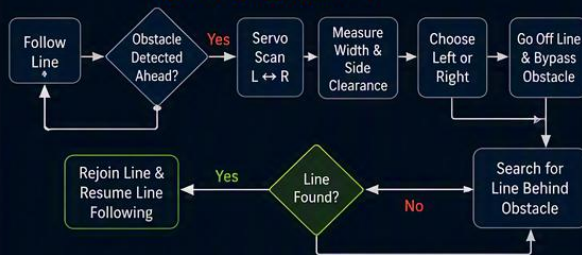
INPUTS

- Ultrasonic distance readings
- Servo scan angle
- Line position from line following system

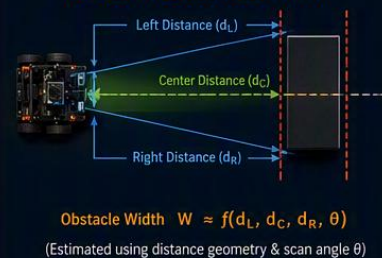
OUTPUTS

- Obstacle detected (Yes/No)
- Estimated obstacle width (W)
- Chosen direction (Left/Right)
- Rejoin line command

HOW IT WORKS (ALGORITHM)



MEASURING OBSTACLE WIDTH



KEY BENEFITS

- Reliable detection of obstacles ahead
- Measures obstacle width
- Intelligent left/right decision
- Safely bypass and rejoin the line
- Maintains mission continuity

WEEK 10 DEMONSTRATION METRICS

TEST	A (Excellent)	B (Good)	C (Satisfactory)	D (Developing)	F (Needs Work)
Obstacle detection rate	100%	≥ 95%	≥ 90%	≥ 80%	< 80%
Width estimation error	≤ 10%	≤ 20%	≤ 30%	≤ 40%	> 40%
Direction selection success	100%	≥ 95%	≥ 90%	≥ 80%	< 80%
Successful line reacquisition	100%	≥ 95%	≥ 90%	≥ 80%	< 80%
Time to bypass obstacle	< 3 s	< 5 s	< 7 s	< 10 s	> 10 s

NOTES

- Adjust speed when near obstacle for safer navigation.
- Use multiple scans for stable width measurement.
- Ensure smooth transitions when leaving and rejoining the line.
- Log events and decisions for debugging and evaluation.



Core work

Coarse scan, then fine scan around obstacles
 Estimate location, width and clearance
 Plan stop, turn or reverse actions
 Recover and reacquire the original line

Expected deliverables

Scanning subsystem
 Obstacle profile generator
 Avoidance and recovery algorithms

What the Team Will Be Assessed On

A fast subsystem is not enough if the integrated robot is unreliable.

Mission

Objectives completed

Navigation

Accuracy and control

Perception

Barcode, hump, obstacle

Robustness

Recovery and stable operation

Engineering

Software quality and efficiency

Integration

Interfaces and teamwork

Demonstration

Clear evidence under test

Start With Interfaces, Then Build Evidence

- 1 Define inputs, outputs and ownership
- 2 Create repeatable subsystem tests
- 3 Record measurements and calibration
- 4 Integrate early through stable APIs
- 5 Practise the Week 10 demonstration

The goal is a dependable integrated robot, supported by measurable engineering evidence.

